

CSE400 – Fundamentals of Probability in Computing

Lecture 23 Scribe (Closed-Notes Exam Revision)

Lecture Title: Predicting the Impossible: Tail Bounds and System Reliability

Case Study: Amazon Black Friday Sale 2026

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1. Case Study Context: Amazon Black Friday Sale 2026

The sale is expected to begin on:

Friday, November 27, 2026

The timeline is inferred from historical patterns and expert forecasts.

1.1 Interpretation

The case study is used to:

- Motivate uncertainty in large-scale systems
- Connect real-world events with computational modeling of uncertainty

2. Core Theme: Uncertainty and Computing

The lecture emphasizes:

“Connect with Uncertainty and Computing”

2.1 Implication

Real-world systems involve:

- Uncertain demand
- Random fluctuations

Mathematical tools are required to:

- Quantify uncertainty
- Predict extreme events

3. Tail Bounds Introduced

The lecture introduces three fundamental tools:

- Markov's Inequality
- Chebyshev's Inequality
- Chernoff Bound

These are used to bound probabilities of extreme deviations.

4. Markov's Inequality

4.1 Assumptions

- $X \geq 0$ (non-negative random variable)
- $a > 0$

4.2 Definition

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

4.3 Step-by-Step Reasoning

1. Consider non-negative random variable X
2. Focus on probability $P(X \geq a)$
3. Use expectation $\mathbb{E}[X]$
4. Since $X \geq a$ contributes at least a :

$$\mathbb{E}[X] \geq a \cdot P(X \geq a)$$

5. Rearranging:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

4.4 Conclusion

Provides an upper bound using only expectation.

5. Chebyshev's Inequality

5.1 Assumptions

Random variable X with:

- Mean μ
- Variance σ^2

5.2 Definition

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

5.3 Step-by-Step Reasoning

1. Consider deviation $|X - \mu|$
2. Square both sides:

$$(X - \mu)^2 \geq k^2\sigma^2$$

3. Apply Markov's inequality:

$$P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2\sigma^2}$$

4. Use:

$$\mathbb{E}[(X - \mu)^2] = \sigma^2$$

5. Substitute:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

5.4 Conclusion

Improves over Markov by incorporating variance.

6. Chernoff Bound

6.1 Purpose

Provides tighter (exponentially decreasing) bounds.

6.2 General Form

$$P(X \geq (1 + \delta)\mu) \leq e^{-f(\delta, \mu)}$$

6.3 Step-by-Step Reasoning

1. Start with:

$$P(X \geq a)$$

2. Apply exponential transformation:

$$P(e^{tX} \geq e^{ta})$$

3. Apply Markov's inequality:

$$P(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

4. Choose parameter t to minimize the bound

5. Result yields exponential decay

6.4 Conclusion

- Strongest among the three bounds
- Useful for rare event analysis

7. Conceptual Integration with Case Study

The Amazon sale represents a high-uncertainty system.

7.1 Application of Tail Bounds

- Predict extreme load conditions
- Ensure system reliability

7.2 Logical Flow

- Real-world uncertainty introduced
- Mathematical tools presented
- Applied to system reliability reasoning

8. Class Participation

The lecture includes class participation.

8.1 Interpretation

Students are expected to:

- Engage with concepts
- Apply inequalities in problem-solving

9. Final Summary (Exam-Oriented)

The lecture connects probability theory with system reliability.

9.1 Core Tools

- Markov's Inequality — expectation-based bound
- Chebyshev's Inequality — variance-based bound
- Chernoff Bound — exponential tail bound

9.2 Applications

- Analyze rare events
- Model uncertainty in computing systems
- Support reliability in large-scale applications

End of Lecture 23 Scribe